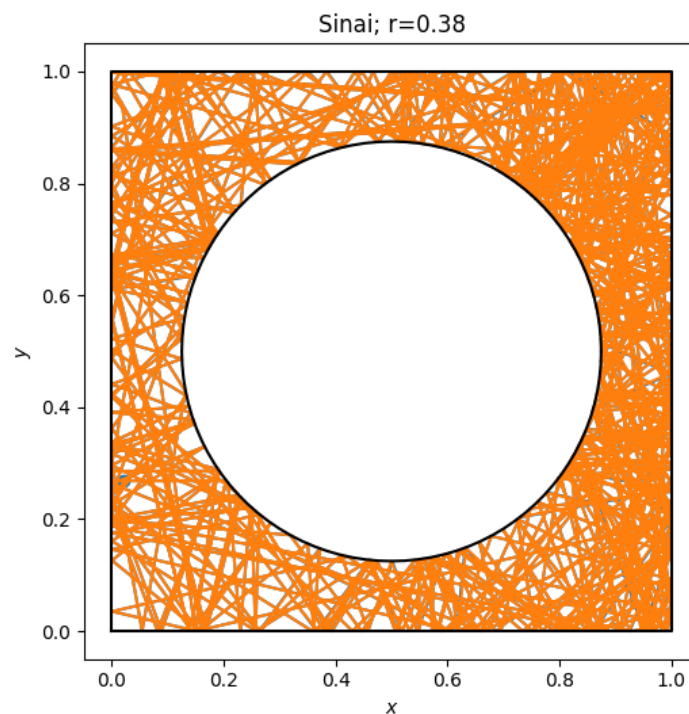


Teorija dinamičnih sistemov



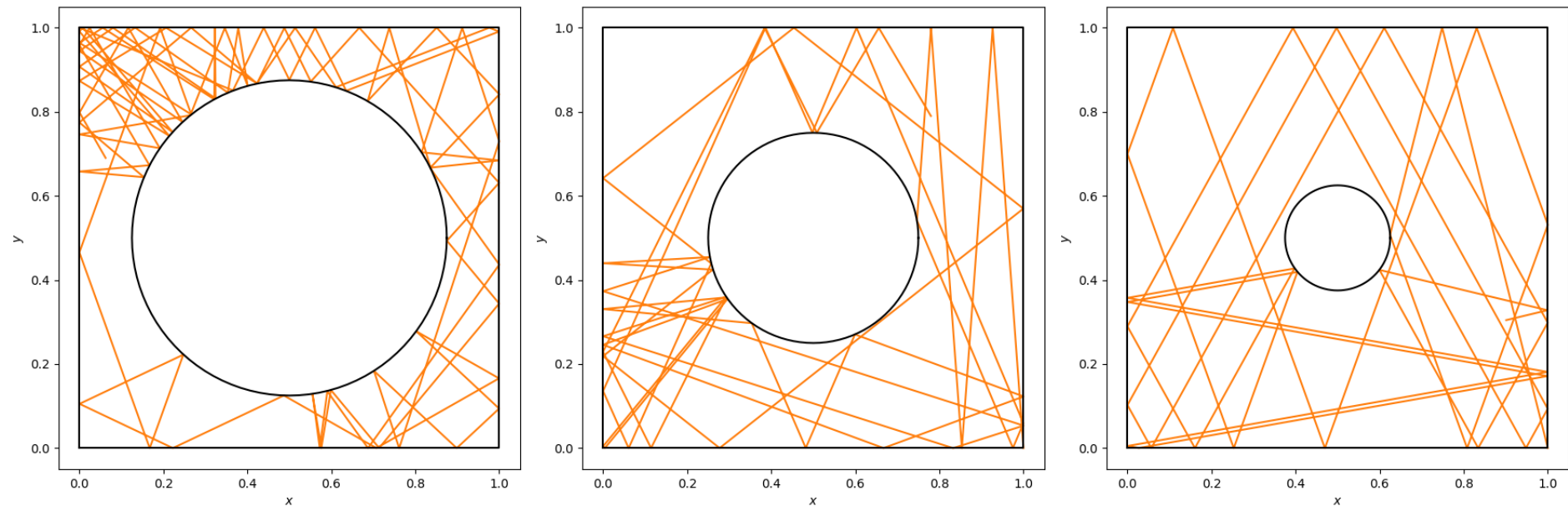
Naloga 25: Sinajev biljard

Žan Butala

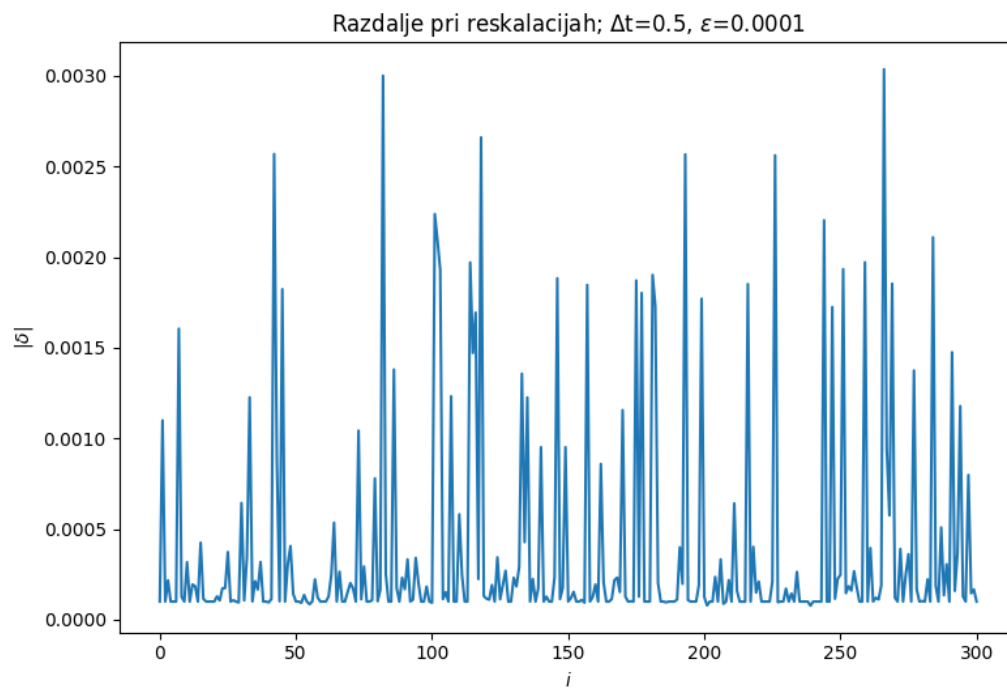
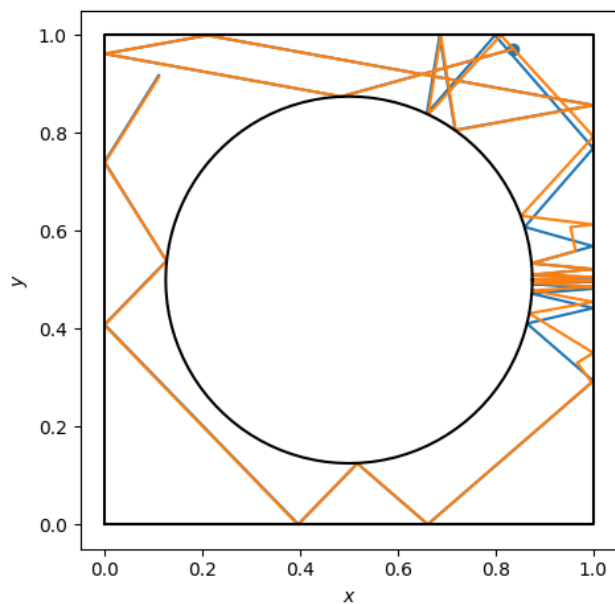
18.1.2024

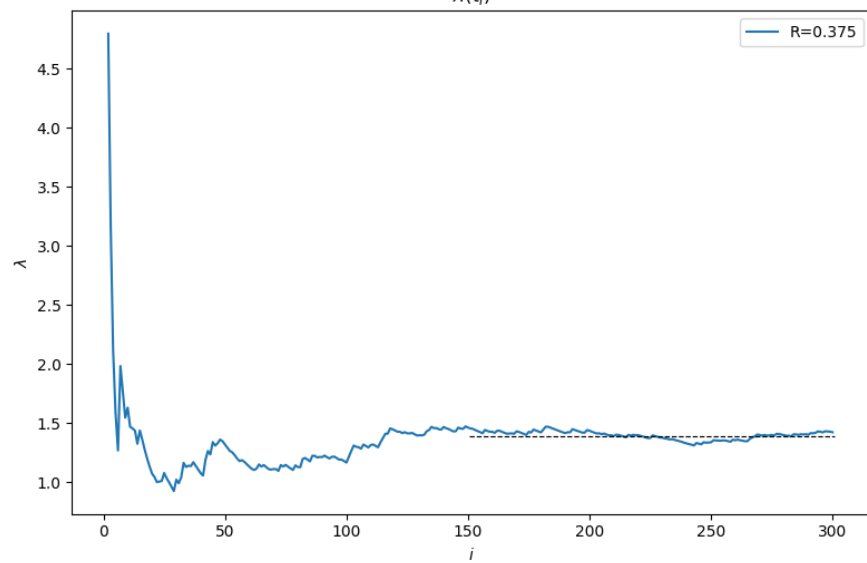
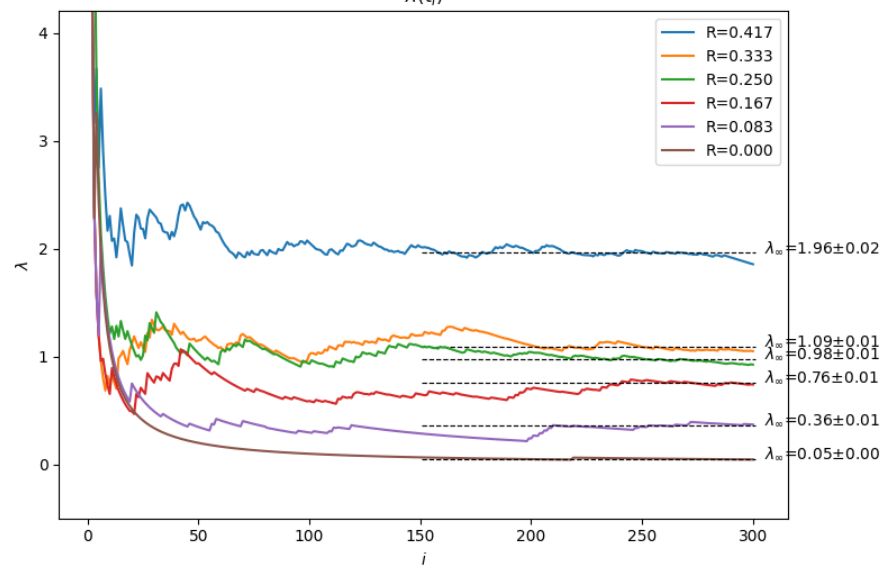
Naloga:

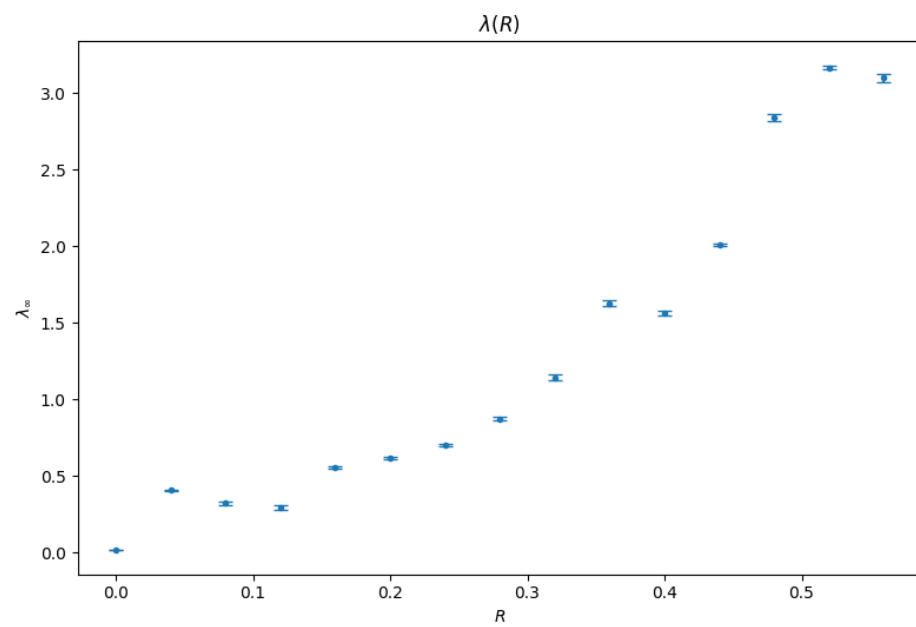
- Izračunati Ljapunove eksponente v odvisnosti od polmera kroga
- Analizirati stabilnost periodičnih orbit



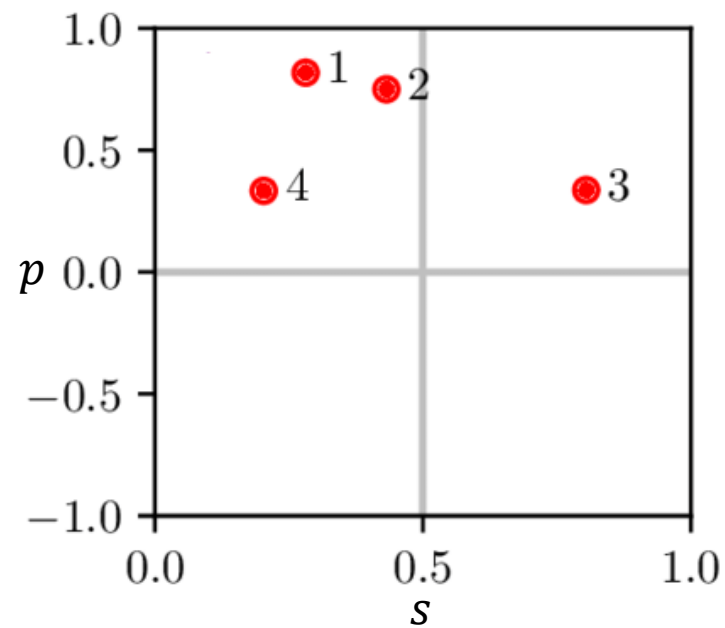
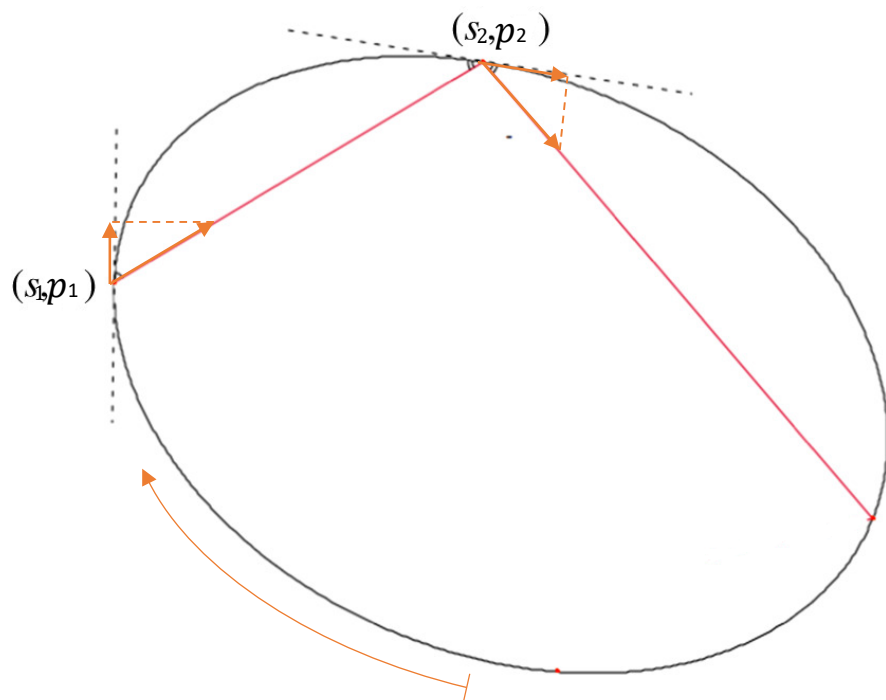
Ljapunov eksponent



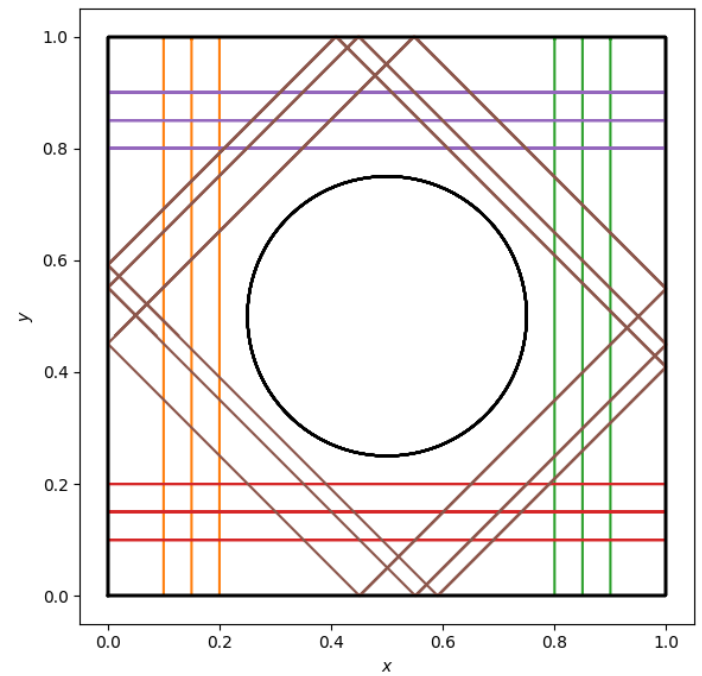
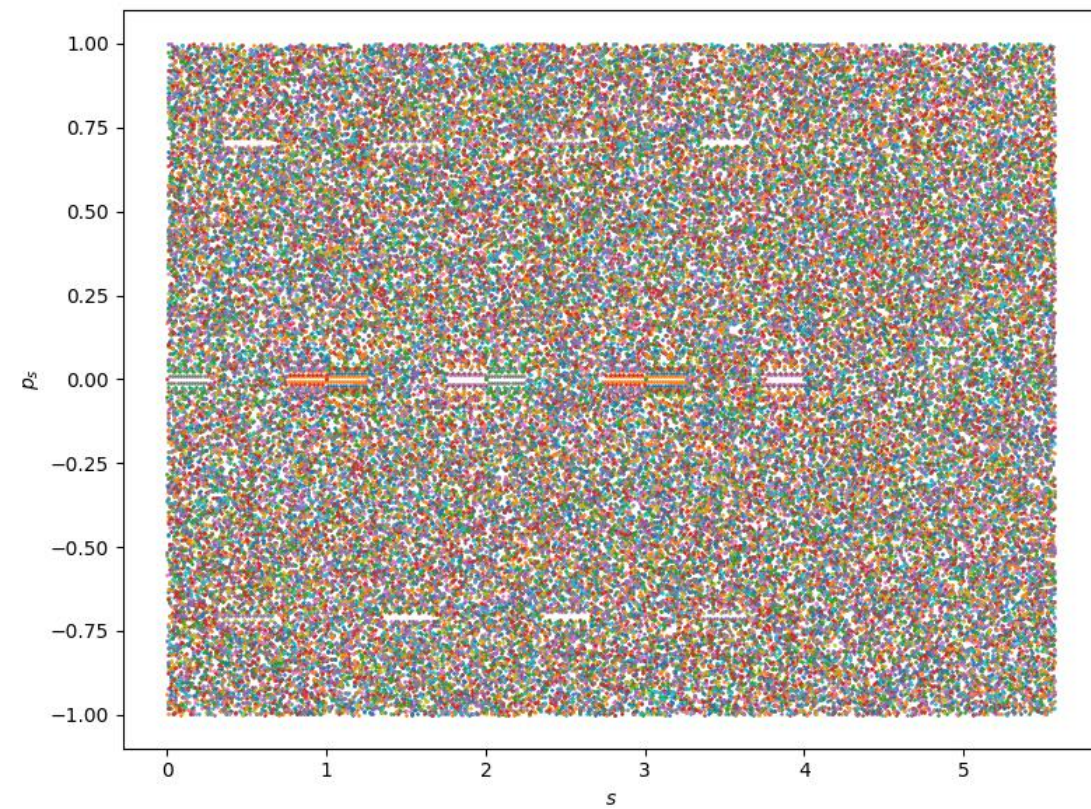
$\lambda(t_i)$  $\lambda(t_i)$ 



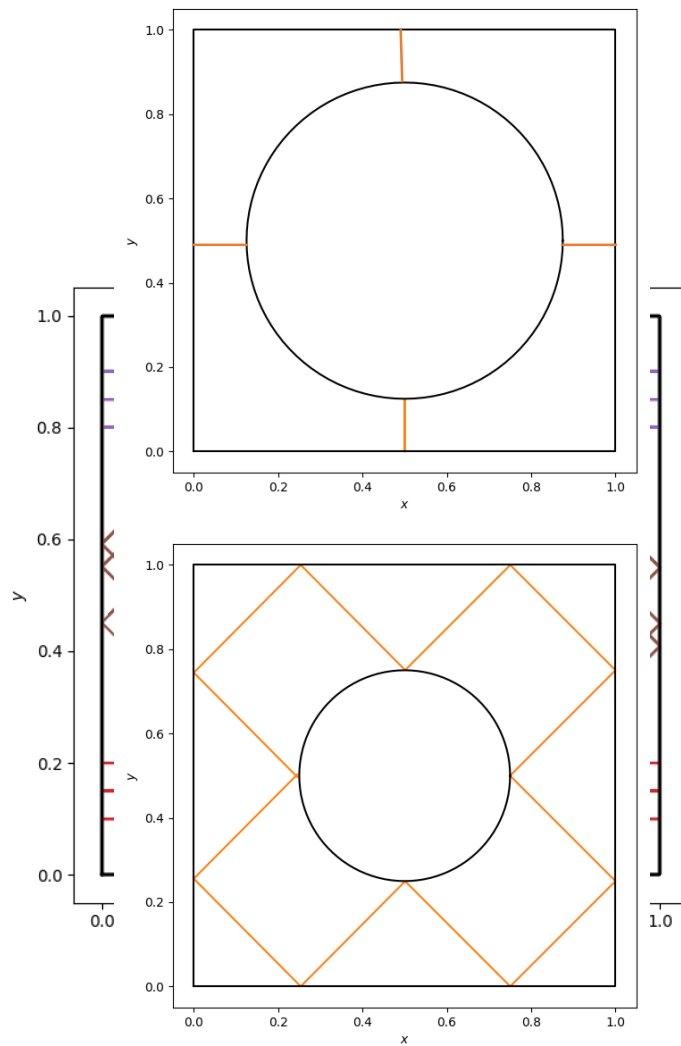
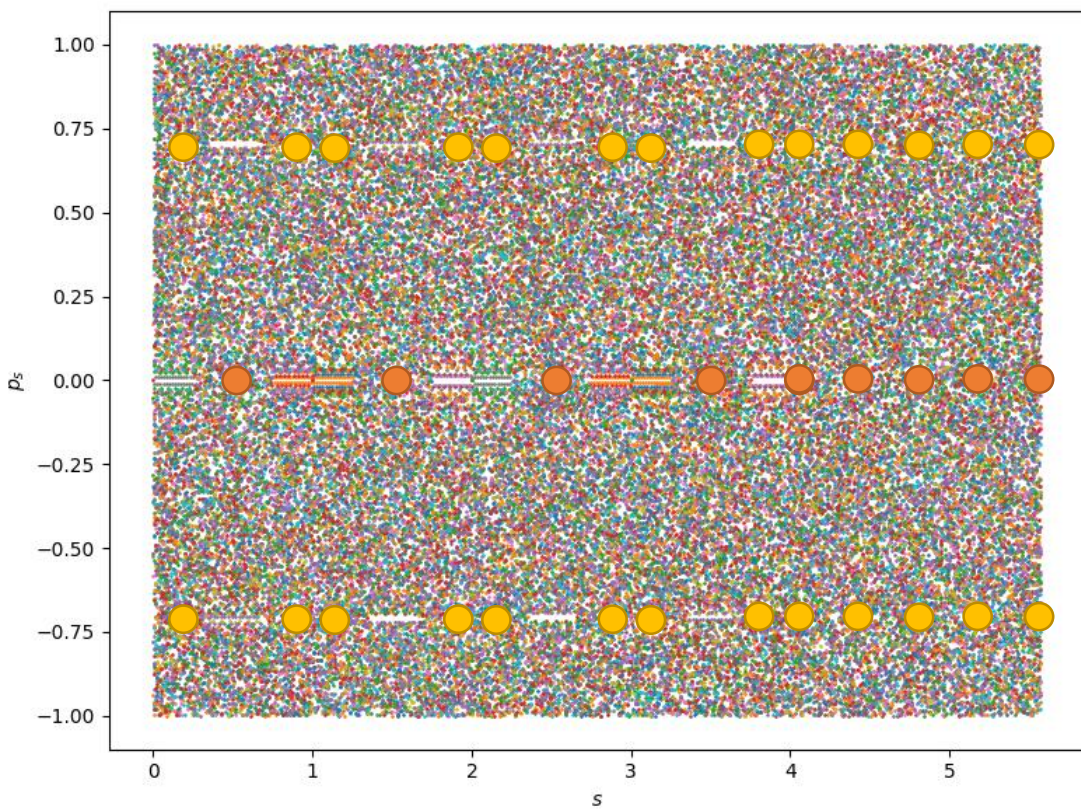
Fazni prostor v P-B koordinatah



Fazni prostor v P-B koordinatah

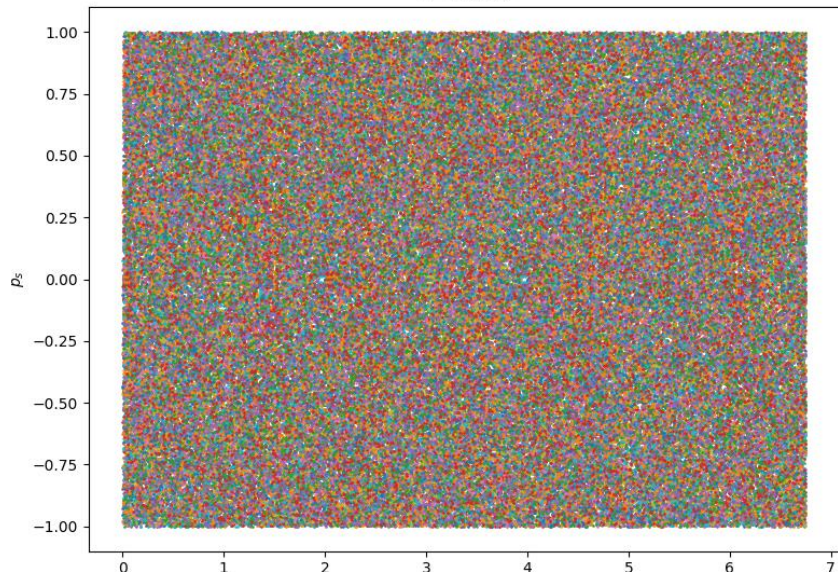


Fazni prostor v P-B koordinatah

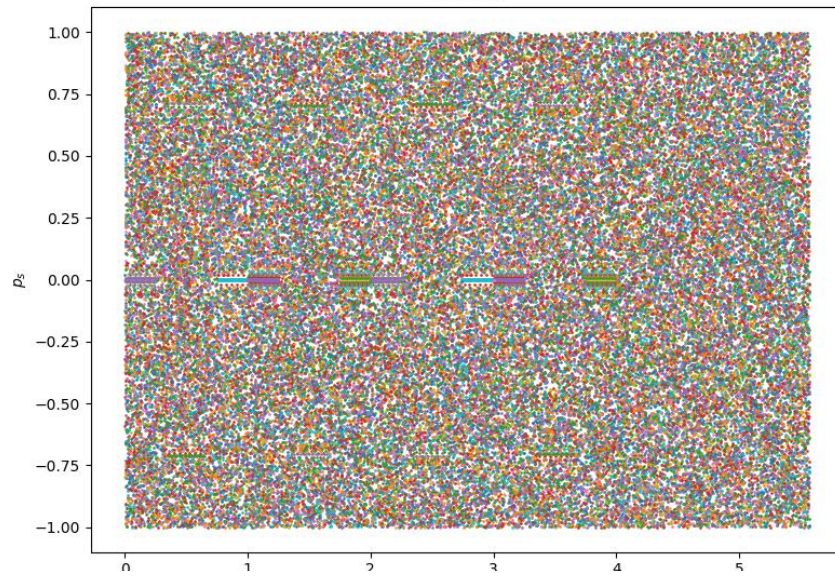


Fazni prostor v P-B koordinatah

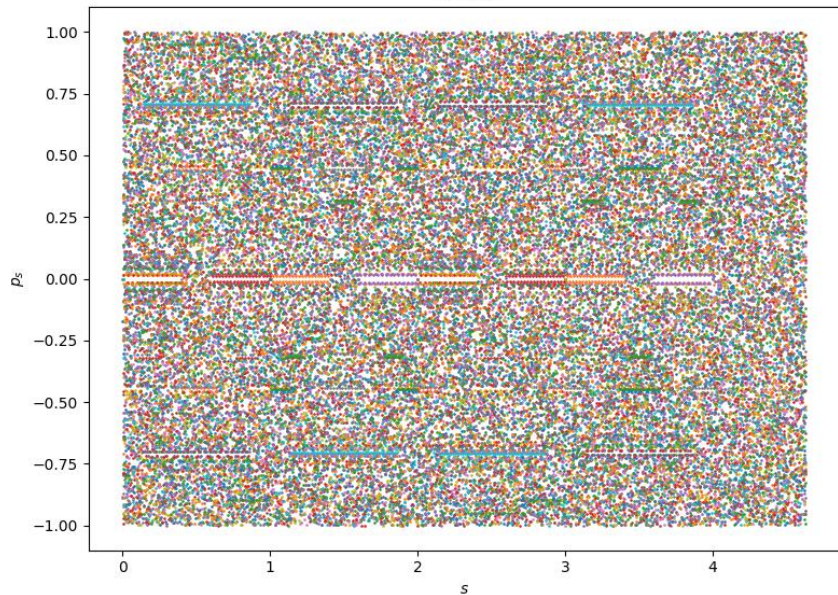
$R = 0.4375$



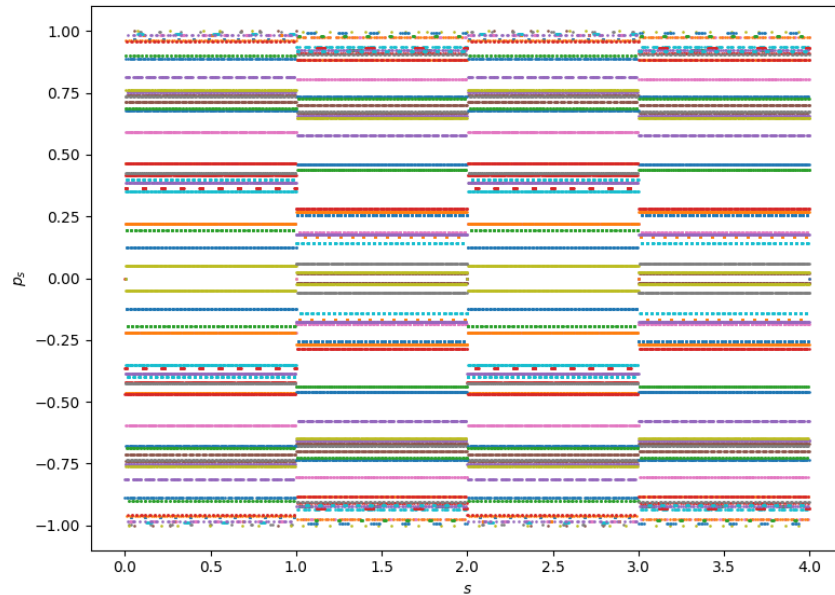
$R = 0.25$



$R = 0.1$



$R = 0.0$



Stabilnost ($z, \theta \ll 1$):

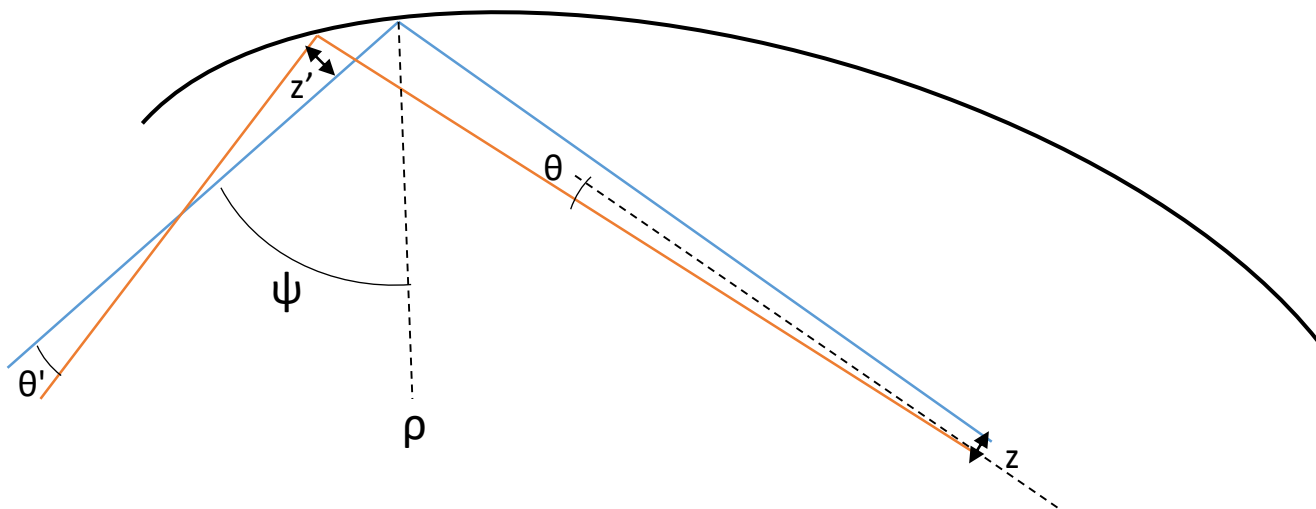
$$\begin{bmatrix} z' \\ \theta' \end{bmatrix} = F_{odboj} F_{prost} \begin{bmatrix} z \\ \theta \end{bmatrix}$$

$$F_{odboj} = \begin{bmatrix} -1 & 0 \\ \frac{2}{\rho \cos \psi} & -1 \end{bmatrix}$$

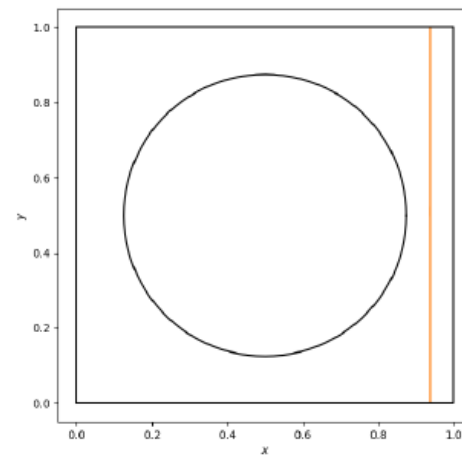
$$F_{prost} = \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix}$$

Pogoj za stabilnost:

$$-2 \leq \text{tr} F \leq 2$$



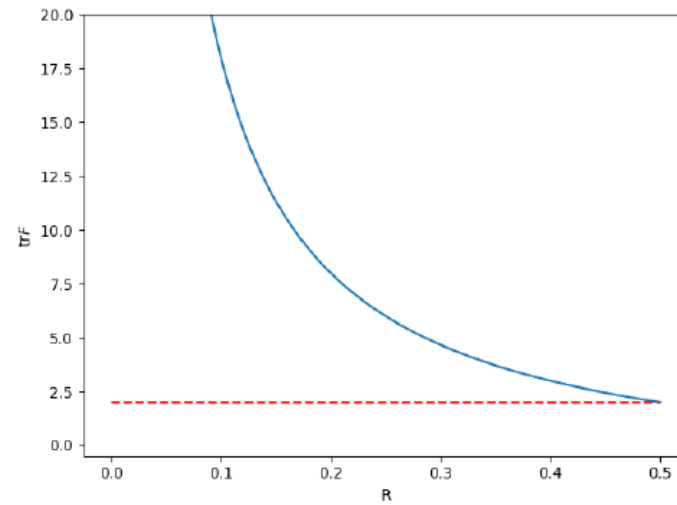
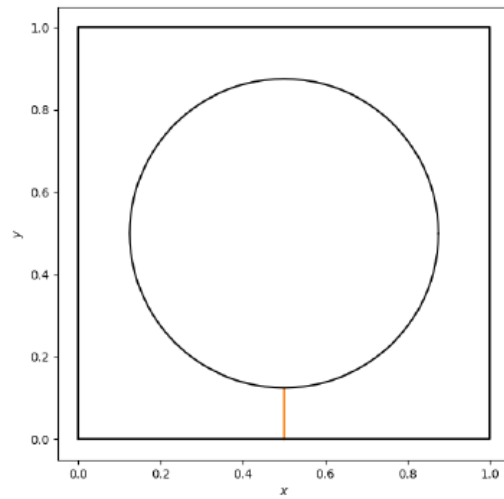
- $\psi_{1,2} = 0, \rho_{1,2} = \infty$



$$\begin{aligned} F &= F_{odb_2} F_{prost_2} F_{odb_1} F_{prost_1} \\ &= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2l \\ 0 & 1 \end{bmatrix} \end{aligned}$$

$$\text{tr} F = 2$$

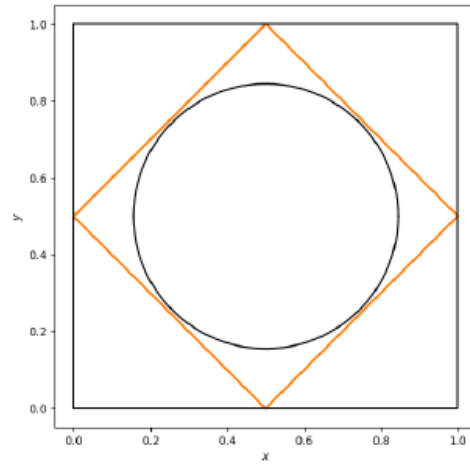
- $\psi = 0, \rho_2 = -R$



$$\begin{aligned}
 F &= F_{odb_2} F_{prost_2} F_{odb_1} F_{prost_1} \\
 &= \begin{bmatrix} -1 & 0 \\ \frac{2}{-R} & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 2l \\ \frac{2}{R} & 1 + \frac{4l}{R} \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 - 2R \\ \frac{2}{R} & 1 + \frac{2}{R} - 4 \end{bmatrix}
 \end{aligned}$$

$$\text{tr} F = -2\left(1 - \frac{1}{R}\right)$$

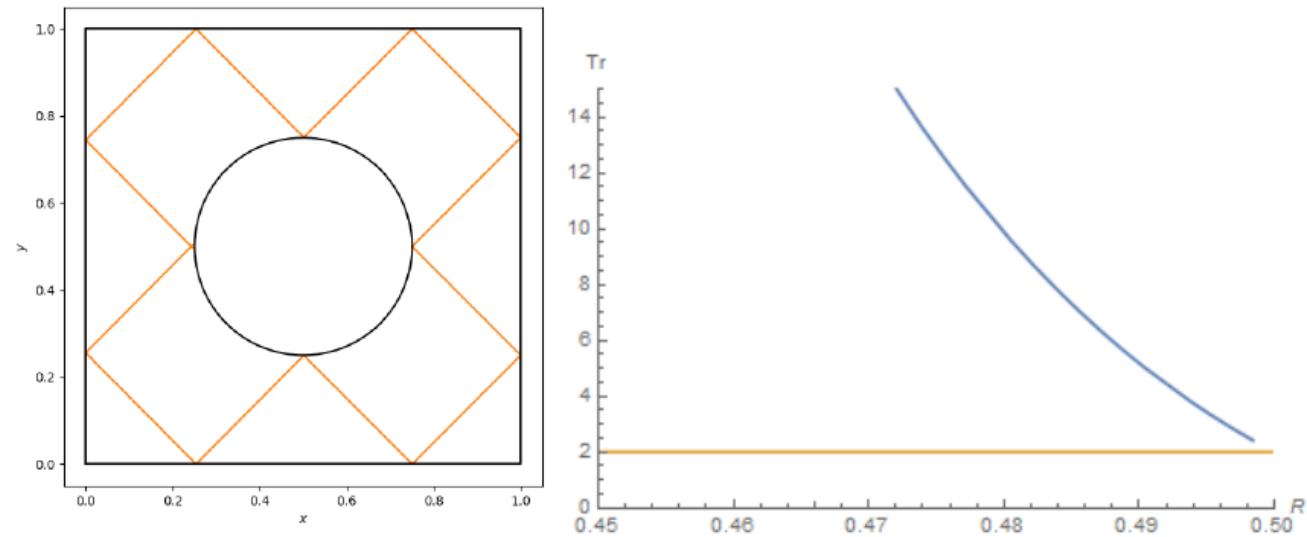
$$\bullet \quad \psi_{1,2,3,4} = \frac{\pi}{4}, \, \rho_{1,2,3,4} = \infty, \, R < \frac{\sqrt{2}}{4}$$



$$\begin{aligned} F &= F_{odb_4} F_{prost_4} F_{odb_3} F_{prost_3} F_{odb_2} F_{prost_2} F_{odb_1} F_{prost_1} \\ &= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & l \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 4l \\ 0 & 1 \end{bmatrix} \end{aligned}$$

$$\mathrm{tr} F = 2$$

- $\psi_i = \frac{\pi}{4}, \rho \in \{-R, \infty\}$



$$\begin{aligned} F &= F_{odb_12} F_{prost_{12}} \cdots F_{odb_2} F_{prost_2} F_{odb_1} F_{prost_1} \\ &= \dots Mathematica \end{aligned}$$

$$\text{tr} F > 2$$

$$l_1 = \sqrt{(1/2 - R)^2 + (1/2 - R)^2};$$

$$l_2 = \sqrt{(1/2 - (1/2 - R))^2 + (1/2 - (1/2 - R))^2};$$

$$\text{MatrixForm}[\text{MatrixPower}[\text{Dot}[A_1, \text{Dot}[B_1, \text{Dot}[A_2, \text{Dot}[B_2, \text{Dot}[A_2, \text{Dot}[B_1]]]]]], 4]]$$

$$\left(\left(1+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}\sqrt{2}\sqrt{R^2}\right)}{R}\right)^2+\left(-\frac{2\sqrt{2}}{R}+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}\sqrt{2}\sqrt{R^2}\right)}{R}\right)\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}+\sqrt{2}\sqrt{R^2}+\left(-2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}-\sqrt{2}\sqrt{R^2}\right)\left(-1+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}\sqrt{2}\sqrt{R^2}\right)}{R}\right)\right)\right)\left(1+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}\sqrt{2}\sqrt{R^2}\right)}{R}\right)\left(-\frac{2\sqrt{2}}{R}+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}\sqrt{2}\sqrt{R^2}\right)}{R}\right)+\left(-\frac{2\sqrt{2}}{R}+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}\sqrt{2}\sqrt{R^2}\right)}{R}\right)\left(\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}\sqrt{2}\sqrt{R^2}\right)}{R}+\left(-1+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}\sqrt{2}\sqrt{R^2}\right)}{R}\right)^2\right)\right)\left(-\frac{2\sqrt{2}}{R}+\frac{2\sqrt{2}\left(2\sqrt{2}\sqrt{\left(\frac{1}{2}-R\right)^2}\sqrt{2}\sqrt{R^2}\right)}{R}\right)$$

$$\text{Simplify}[\text{Tr}[\text{MatrixPower}[\text{Dot}[A_1, \text{Dot}[B_1, \text{Dot}[A_2, \text{Dot}[B_2, \text{Dot}[A_2, \text{Dot}[B_1]]]]]], 4]]]$$

$$\frac{2\left(6049R^4+32R^2\left(125+32\sqrt{(-1+2R)^2}\right)-32R^3\left(244+105\sqrt{R^2}+57\sqrt{(-1+2R)^2}\right)-256R\left(4+\sqrt{(-1+2R)^2}+\sqrt{R^2}\left(3+8\sqrt{(-1+2R)^2}\right)\right)+64\left(2+8\sqrt{R^2}\sqrt{(-1+2R)^2}+3\left(R^2\right)^{3/2}\left(16+15\sqrt{(-1+2R)^2}\right)\right)\right)}{R^4}$$

$$\text{Plot}[\{\%71, 2\}, \{R, 0, 5\}, \text{PlotRange} \rightarrow \{\{0.45, 0.5\}, \{0, 15\}\}, \text{AxesLabel} \rightarrow \{\text{HoldForm}[R], \text{HoldForm}[\text{Tr}]\}]$$

